

The Effect of Elbow and Shoulder Supports on Detection Threshold of Hand Redirection

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ABSTRACT

Hand Redirection (HR) techniques subtly alter a user's real hand movement to deviate from its virtual representation, enhancing passive haptics in virtual environments. However, noticeable discrepancies can reduce immersion. Inspired by redirected walking research demonstrating that knee supports can influence the perception of manipulation, this study explores whether shoulder and elbow supports can expand the detection threshold in horizontal HR. A user study revealed that shoulder supports significantly increase the DT, offering a practical, stimulus-free method to enhance HR. These findings highlight the potential of leveraging joint constraints to improve immersion without relying on external stimuli.

Index Terms: virtual reality, hand redirection, detection threshold

1 INTRODUCTION

Passive haptics, a straightforward method for providing haptic feedback in VR, delivers tactile sensations by using physical objects. However, its practicality is limited because it requires a corresponding physical object for each virtual object in the VR environment. To address this issue, a widely studied technique called Hand Redirection (HR) [1] has been developed. By manipulating the real position and orientation of the user's hand and applying it in the VR environment, HR enables a single physical object to provide tactile feedback for multiple virtual objects.

To utilize HR more widely, it is necessary to expand its Detection Threshold (DT)—the maximum amount of manipulation that goes unnoticed by the user. Zenner Krüger [6] proposed a method to determine how much to manipulate the user's hand and investigated the DT using that approach, reporting a DT of 8.90° for horizontal redirection. Similarly, we measured the DT using a comparable method in this study. Previous attempts to expand the DT have included introducing proprioceptive noise via electrical stimuli [4]. However, such approaches often diminish immersion due to reliance on external stimuli.

In contrast, research on Redirected Walking (RDW) has shown that wearing knee supports can influence RDW's applicability without requiring dynamic manipulation [3]. Inspired by this finding, we investigated whether attaching supports to the shoulder and elbow joints could also expand the DT for HR. This study examines the feasibility of a simpler and more effective method to enhance the DT for HR without compromising immersion.

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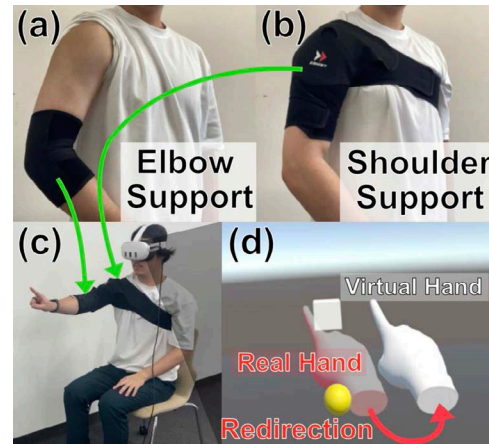


Figure 1: Examining the Effects of (a) Elbow and (b) Shoulder Supports on Detection Threshold for Hand Redirection in VR. The real hand's position (red hand) is redirected to a virtual hand (white hand) during a reaching task.

2 USER STUDY

The purpose of this experiment was to investigate whether wearing supports on the elbow and shoulder expands the DT for horizontal HR. We conducted a within-participants experiment with two factors: shoulder support and elbow support, resulting in four conditions. The experiment was approved by the Ethics Committee of the Graduate School of Information Science and Technology at the University of Tokyo (UT-IST-RE-240205-3).

2.1 Apparatus

In this experiment, we used two types of supports as shown in Figure 1: (a) elbow supports (M, L, LL, 3L) and (b) shoulder supports (M, L, LL), both manufactured by ZAMST. The size used for each participant was determined based on the circumference of the upper arm and shoulder joint. Participants wore a Meta Quest 3 head-mounted display (HMD) during the trials.

2.2 Conditions and Procedure

Twenty-four healthy participants (10 females, 14 males), aged between 21 and 57 years ($M = 27.1$, $SD = 9.52$), took part in the experiment. None had visual or arm-movement impairments. We used a within-subjects design with two factors (shoulder and elbow support), creating four conditions: "shoulder," "elbow," "shoulder $\times$ elbow," and "control."

For redirection, we applied Zenner & Krüger's method [6], testing nine angles ($0^\circ, \pm 4^\circ, \pm 8^\circ, \pm 12^\circ, \pm 16^\circ$) ten times each, totaling 360 trials. Each support condition was conducted in a separate block, with the order of conditions and angles randomized.

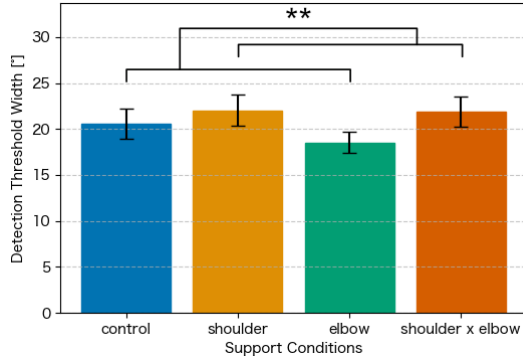


Figure 2: Mean detection thresholds width. The error bars represent the standard error.

Before the trials, participants completed a brief demographic questionnaire (age, height, handedness). We measured each participant's upper arm and shoulder circumferences to determine appropriate support sizes. Afterward, participants wore the HMD and the required supports, and began the main trials.

During each trial, participants were instructed to touch a spherical object (Figure 1, (d)) which indicates the start point positioned 30 cm in front of and 30 cm below the HMD with their right index finger. They then extended their hand straight forward to touch a white cubic object (Figure 1, (d))) located 40 cm ahead. After each trial, participants answered a question: "Did the movement of the virtual hand match your real hand? (Yes/No)"

Upon completing all trials, participants removed the equipment and answered a brief oral interview.

3 RESULTS AND DISCUSSION

Table 1: A table of mean values of left DT, right DT, and DT width for each condition (mean $\pm$ SE).

condition	Left DT [°]	Right DT [°]	DT Width [°]
control	-10.5 $\pm$ 1.0	10.1 $\pm$ 0.9	20.6 $\pm$ 1.6
shoulder	-10.9 $\pm$ 1.0	11.1 $\pm$ 0.9	22.0 $\pm$ 1.7
elbow	-9.59 $\pm$ 0.9	8.93 $\pm$ 0.9	18.5 $\pm$ 1.1
shoulder $\times$ elbow	-11.0 $\pm$ 1.0	11.0 $\pm$ 0.9	21.9 $\pm$ 1.7

Table 1 presents the mean values of left DT, right DT, and DT width, and Figure 2 shows the mean DT width for each condition across all participants. The DT was calculated using Psignifit 4 [5].

For each participant, we plotted the manipulation angle (α) on the horizontal axis and the percentage of "No" responses on the vertical axis. Fitting was performed separately for leftward angles ($\alpha \leq 0$) and rightward angles ($\alpha \geq 0$), using an inverse sigmoid function for leftward angles and a standard sigmoid function for rightward angles. The angles where the "No" response rate reached 0.5 were referred to as the left DT for the leftward direction and the right DT for the rightward direction. The DT width was defined as the difference between the right DT and left DT.

A Shapiro-Wilk test confirmed the normality of the results for all conditions. A two-way repeated measures ANOVA, with shoulder and elbow supports as independent variables and DT width as the dependent variable, revealed a significant main effect of shoulder support ($F = 10.36, p < 0.01$). However, there was no significant main effect of elbow support ($F = 3.45, p = 0.076$) or interaction effect ($F = 3.57, p = 0.076$).

The mean DT width for conditions with shoulder support (shoulder, shoulder $\times$ elbow) was $22.0 \pm 1.2^\circ$ (mean $\pm$ SE), compared

to $19.5 \pm 1.0^\circ$ for conditions without shoulder support (control, elbow), representing a 13% increase. This effect is comparable to the 11% increase observed by Ogawa et al. with electrical noise stimulation [4], suggesting that shoulder supports can expand the DT to a similar degree as external stimulation while preserving immersion.

Post-experiment interviews reinforce these findings, with 12 of 24 participants noting the impact of shoulder supports, compared to only 2 mentioning elbow supports. The shoulder, as a larger and more mobile joint, plays a dominant role in movement generation and control, while the elbow stabilizes the forearm's trajectory near the endpoint [2]. Shoulder constraints may stabilize large movements, whereas elbow constraints could disrupt fine adjustments, increasing discomfort during virtual hand displacement.

4 CONCLUSION AND FUTURE WORK

This study investigated the impact of shoulder and elbow supports on the detection threshold (DT) of hand redirection (HR). The results demonstrated a significant effect of shoulder supports, with a notable increase in DT. In contrast, no significant effects were found for elbow supports.

Unlike previous methods, shoulder supports eliminate the need for external stimuli, which may disrupt immersion and require external power sources. Shoulder supports involve fewer technical limitations, offering a more practical solution for expanding DT.

Since the precise mechanisms by which shoulder supports expand DT remain unclear, further research is needed to clarify their influence on proprioception and broader applications. Specifically, future studies should examine arm movements in more detail (including vertical and depth directions) to better understand these effects. Furthermore, combining shoulder supports with existing techniques such as electrical stimulation could further expand the DT of HR.

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